Practical No. 9

Aim

To write and execute a program to calculate **page hits and page faults using the Optimal Page Replacement algorithm**.

Step 1: Open Git Bash

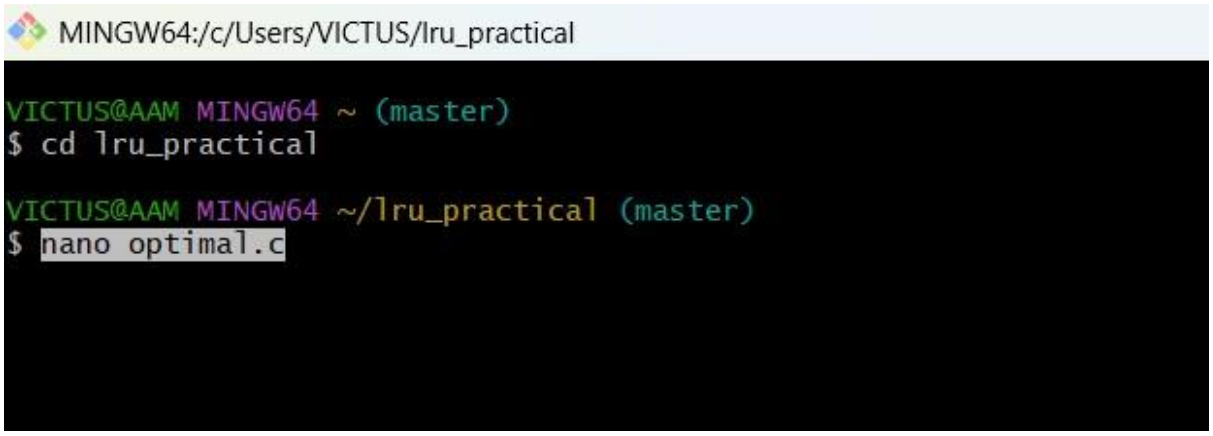
Git Bash was opened to execute commands.

Step 2: Navigate to Directory

The previously created directory was opened:

Step 3: Create C File

A new C file was created using nano editor:

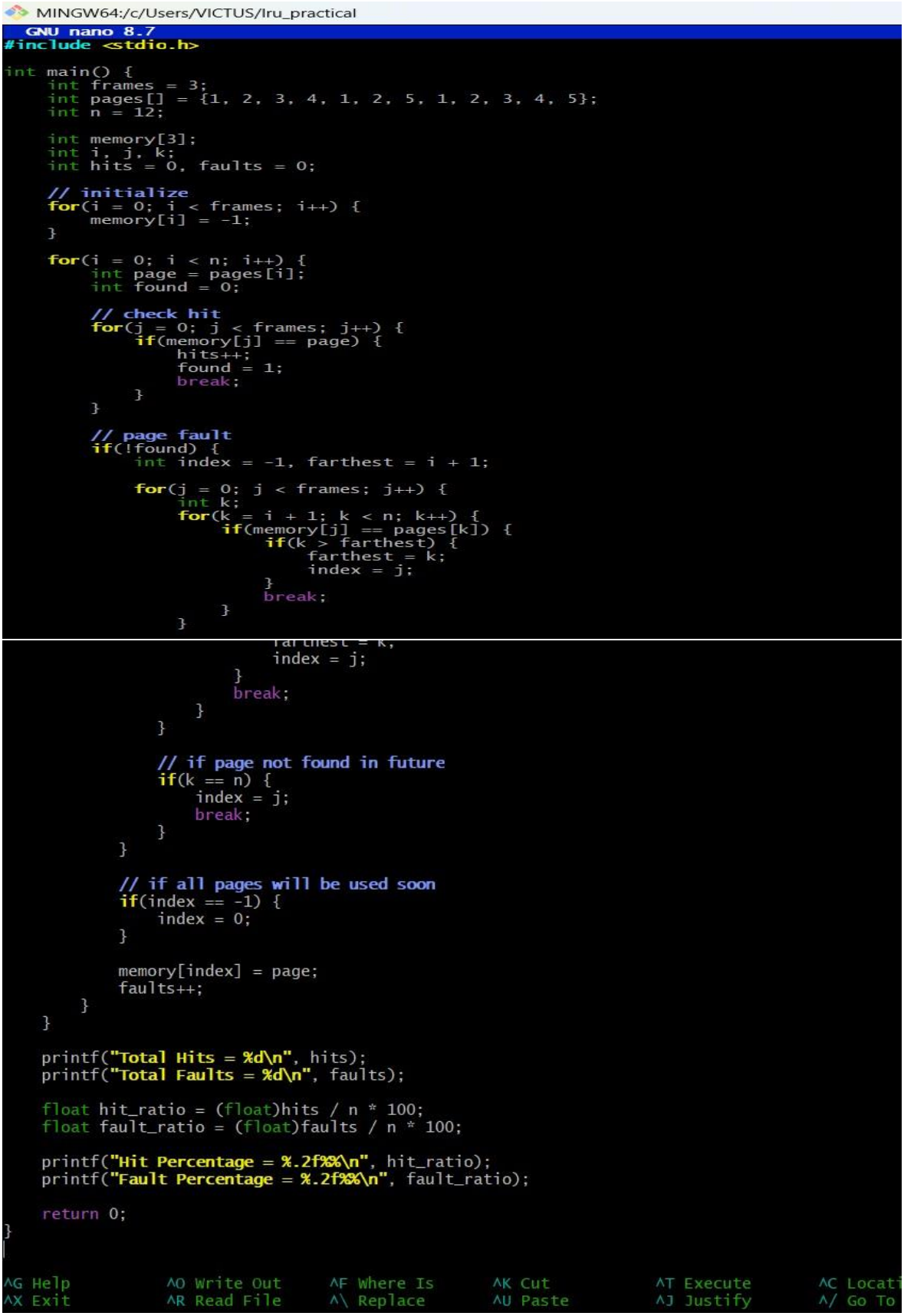


Step 4: Write Program

The Optimal Page Replacement program was written in C language with:

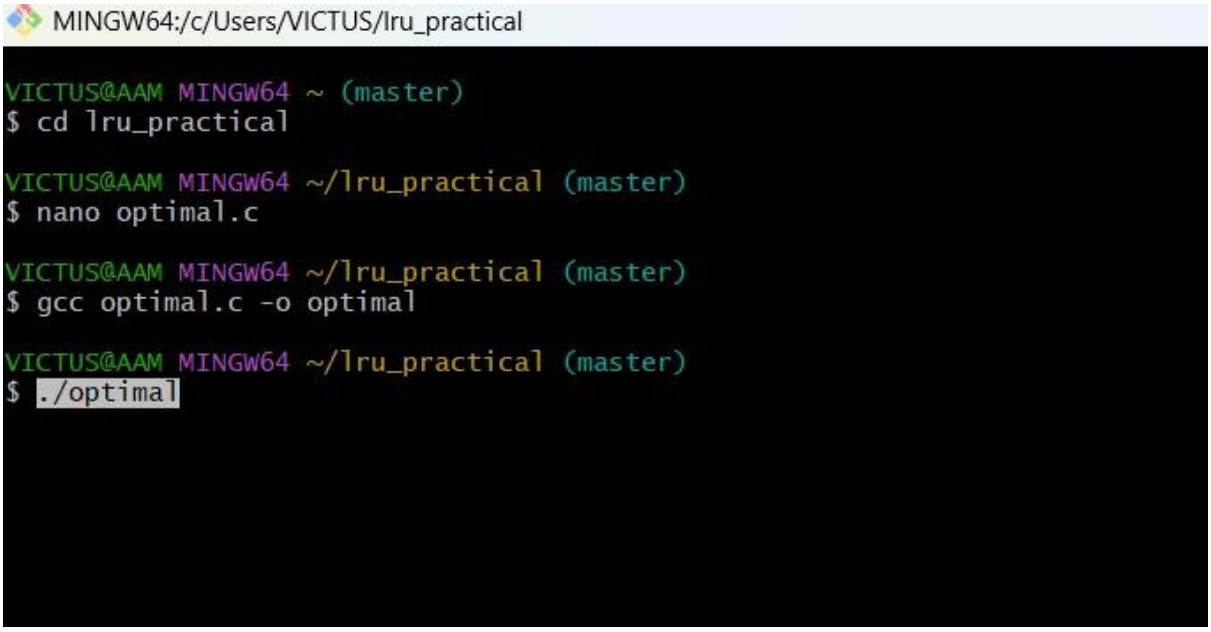
- Number of frames = 3
- Page reference string = 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5

  The program calculates total page hits and page faults based on future page usage.



Step 5: Compile and Execute Program

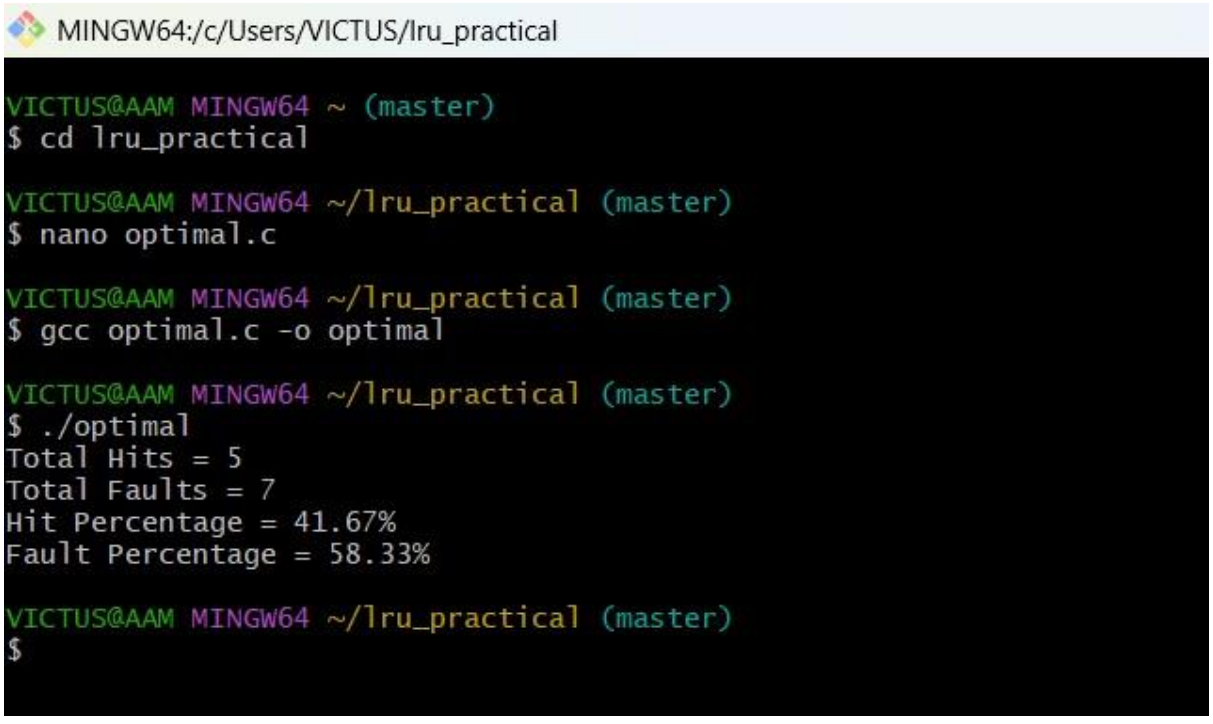
The program was compiled and executed using:



Step 6 : Output Observed

The output obtained is:

- Total Hits = 5
- Total Faults = 7
- Hit Percentage = 41.67%
- Fault Percentage = 58.33%



Result

The Optimal Page Replacement program was successfully executed and the page hits and page faults were calculated correctly.